Topic: Pokemon Trainer Move Selection AI (Turn-based)

Deadline 19th June

Algorithms

Simultaneous Move Alpha-Beta Pruning (SMAB) is chosen to bypass Standard Minimax with Alpha-Beta Pruning. This is because Pokémon turns occur simultaneously but Standard Minimax with Alpha-Beta Pruning relies on alternating, sequential turns where one player minimizes while the other maximizes. This causes a standard minmax tree to suffer from structural inseparability. Whereas for SMAB, every decision node represents a zero-sum payoff matrix mapping the agent's choices against the opponent choices. The algorithm computes upper and lower bounds for achievable utilities, utilizing linear programming or strict dominance checks to prune branches where an action is mathematically dominated by an opponent's mixed counterstrategy.

Information Set Monte Carlo Tree Search (IS-MCTS) helps to navigate Pokémon's hidden state elements, like unrevealed enemy items, based abilities, and move pools. Before executing the four-stage MCTS loop, selection, expansion, simulation and backpropagation, the algorithm samples a determinized state instance. This instance populates the opponent's hidden variables based on empirical competitive meta-game usage statistics. The search build statistics over a collection of these plausible determinized game states, balancing exploration and exploitation via a modified Upper Confidence bound applied to the trees metric to converge on the move with the highest robust expected utility.

Goals

- **Mathematical Modification:** To adapt standard sequential SMAB and MCTS into the multi agent matrices capable of executing decisions under simultaneous action selection and hidden variables.
- **Heuristic Reward Function Design:** To construct a comprehensive state-evaluation heuristic function that scores game states by combing raw hit-point percentages with critical strategic variable, specifically accounting for stat modifiers, team type coverage remaining and entry hazards.
- **Empirical Benchmarking:** To conduct a rigorous quantitative evaluation by pitting both experimental agent against a greedy baseline max-damage script across a minimum sample of 500 simulated battles

Description

This project develops the core decision-making architecture of an automated Pokémon Trainer agent and deploy them inside a self-contained, turn based battle environment.

At the initiation of each combat turn, the simulator transmits a detailed state vector containing the player's active attributes, the health pools of all backline teammates, active battlefield status effects and all historical data on the opponent's team. It presents a branching factor consisting of up to 4 active moves or 5 potential team-switching actions. The agent pipes this state vector simultaneously into the SMAB and IS-MCTS modules. The algorithms then construct future projections up to a designed computational token budget, evaluate the minimax paths or rollout win-probabilities across the generated trees and issue the optimal action back to the simulator server for real-time resolution.

Why is this work relevant?

- It breaks the alternating turn assumption where traditional tree-searching algorithms assume perfectly sequential turns. Traditional tree-searching algorithms are structurally incompatible with games defined by simultaneous choices where psychological prediction, bluffing and risk management are mandatory. This solves the problem and ensures that turns can be completed simultaneously.
- Pokémon is a stochastic game containing combinatorial state configurations across hundreds of distinct characters, moves and random number generator variables. This requires statistical optimizations that go beyond trivial brute-force lookups.
- The techniques engineered to solve this problem can be applied to real-world challenges outside of video games, such as algorithmic high-frequency trading under volatile market trends, multi-agent robotic logistics coordination and autonomous navigation inside unpredictable, partially obscured physical spaces.

References

https://gist.github.com/lhearachel/ff61af1f58c84c96592b0b8184dba096

Alpha-Beta Pruning for Games with Simultaneous Moves, https://www.researchgate.net/publication/223073247_Alpha-Beta_Pruning_for_Games_with_Simultaneous_Moves

Information Set Monte Carlo Tree Search, https://eprints.whiterose.ac.uk/id/eprint/75048/1/CowlingPowleyWhitehouse2012.pdf